

# Probability & Statistics: Homework 3 – Fully Organized Solutions

Below each problem I follow the same structure: - **Identify the model** - **Write the key formulas** - **Do the step-by-step calculation** - **State the final answer clearly**

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## Problem 1 – Biased Coin (Binomial)

**Problem.** A biased coin has probability  $p = 0.3$  of landing heads. It is tossed  $n = 10$  times. Let  $X$  be the number of heads.

### (a) Identify the distribution and parameters

- We have **independent, identical trials** (each toss) with two outcomes (H/T) and constant success probability  $p = 0.3$ .
- Therefore

$$X \sim \text{Binomial}(n = 10, p = 0.3).$$

### (b) Write down the pmf

For a binomial random variable:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n.$$

Here:

$$P(X = k) = \binom{10}{k} (0.3)^k (0.7)^{10-k}.$$

### (c) Compute $P(X = 0)$ , $P(X = 3)$ , $P(X \geq 1)$

#### 1. $P(X = 0)$

$$P(X = 0) = \binom{10}{0} (0.3)^0 (0.7)^{10} = (0.7)^{10}.$$

Numerically:

$$(0.7)^{10} \approx 0.02825.$$

#### 2. $P(X = 3)$

$$P(X = 3) = \binom{10}{3} (0.3)^3 (0.7)^7.$$

Compute step by step:  $\binom{10}{3} = 120$  -  $(0.3)^3 = 0.027$  -  $(0.7)^7 \approx 0.08235$

So

$$P(X = 3) \approx 120 \cdot 0.027 \cdot 0.08235 \approx 0.266.$$

3.  $P(X \geq 1)$

Use the complement rule:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.7)^{10} \approx 1 - 0.02825 = 0.97175.$$

#### (d) Mean and variance

For  $X \sim \text{Binomial}(n, p)$ :

$$E[X] = np, \quad \text{Var}(X) = np(1 - p).$$

So here:

$$E[X] = 10(0.3) = 3,$$

$$\text{Var}(X) = 10(0.3)(0.7) = 2.1.$$

**Final summary for Problem 1:** -  $X \sim \text{Binomial}(10, 0.3)$ . -  $P(X = 0) = 0.7^{10} \approx 0.0283$ . -  $P(X = 3) \approx 0.266$ . -  $P(X \geq 1) \approx 0.9718$ . -  $E[X] = 3$ ,  $\text{Var}(X) = 2.1$ .

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## Problem 2 – Defective Bulbs (Binomial)

**Problem.** Each bulb is defective with probability  $p = 0.02$ , independently. A box has  $n = 50$  bulbs. Let  $Y$  be the number of defective bulbs.

### (a) Identify the distribution

We have 50 independent bulbs, each defective (success) with probability 0.02. Thus

$$Y \sim \text{Binomial}(n = 50, p = 0.02).$$

### (b) Compute $P(Y = 0)$

Using the binomial pmf:

$$P(Y = 0) = \binom{50}{0} (0.02)^0 (0.98)^{50} = (0.98)^{50}.$$

Numerically,

$$P(Y = 0) \approx 0.364.$$

**(c) Compute  $P(Y \leq 2)$**

We sum  $k = 0, 1, 2$ :

$$P(Y \leq 2) = P(Y = 0) + P(Y = 1) + P(Y = 2).$$

$$\bullet P(Y = 0) = (0.98)^{50} \approx 0.364.$$

$$\bullet P(Y = 1) = \binom{50}{1} (0.02) (0.98)^{49} = 50(0.02)(0.98)^{49}. \text{ Numerically, this is about } 0.372.$$

$$\bullet P(Y = 2) = \binom{50}{2} (0.02)^2 (0.98)^{48}. \text{ This is approximately } 0.186.$$

So

$$P(Y \leq 2) \approx 0.364 + 0.372 + 0.186 \approx 0.922.$$

**(d) Mean and standard deviation**

For  $Y \sim \text{Binomial}(n, p)$ :

$$E[Y] = np = 50(0.02) = 1.$$

$$\text{Var}(Y) = np(1 - p) = 50(0.02)(0.98) = 0.98.$$

$$\text{SD}(Y) = \sqrt{0.98} \approx 0.99.$$

**Final summary for Problem 2:** -  $Y \sim \text{Binomial}(50, 0.02)$ . -  $P(Y = 0) = 0.98^{50} \approx 0.364$ . -  $P(Y \leq 2) \approx 0.922$ . -  $E[Y] = 1$ ,  $\text{Var}(Y) = 0.98$ ,  $\text{SD}(Y) \approx 0.99$ .

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## Problem 3 – Binomial CDF Interpretation

Let  $Z \sim \text{Binomial}(n = 8, p = 0.6)$ .

**(a) Write  $P(Z \leq 3)$  as a sum**

By definition of the binomial CDF:

$$P(Z \leq 3) = \sum_{k=0}^3 P(Z = k) = \sum_{k=0}^3 \binom{8}{k} (0.6)^k (0.4)^{8-k}.$$

**(b) Verbal interpretation of  $Z \leq 3$**

"At most 3 successes in 8 trials" – i.e. in 8 independent trials with success probability 0.6, there are 0, 1, 2, or 3 successes.

**(c) Meaning of  $P(Z \leq k) = 0.9$**

If  $P(Z \leq k) = 0.9$ , this means: - 90% of the probability mass of  $Z$  lies at or below  $k$ . - Equivalently,  $k$  is the **90th percentile** (or a 90% quantile) of the binomial distribution with parameters  $n = 8, p = 0.6$ .

## Problem 4 – Poisson Distribution (Calls per Hour)

**Problem.** Calls arrive with rate  $\lambda = 4$  per hour. Let  $N$  be the number of calls in one hour.

**(a) Identify distribution and pmf**

$$N \sim \text{Poisson}(\lambda = 4).$$

The Poisson pmf is

$$P(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

So specifically

$$P(N = k) = e^{-4} \frac{4^k}{k!}.$$

**(b) Compute  $P(N = 0), P(N = 1), P(N \geq 2)$**

1.  $P(N = 0)$ :

$$P(N = 0) = e^{-4} \frac{4^0}{0!} = e^{-4} \approx 0.0183.$$

2.  $P(N = 1)$ :

$$P(N = 1) = e^{-4} \frac{4^1}{1!} = 4e^{-4} \approx 0.0733.$$

3.  $P(N \geq 2)$ : Use the complement:

$$P(N \geq 2) = 1 - P(N = 0) - P(N = 1) \approx 1 - 0.0183 - 0.0733 \approx 0.9084.$$

### (c) Mean and variance

For  $\text{Poisson}(\lambda)$ :

$$E[N] = \lambda, \quad \text{Var}(N) = \lambda.$$

So here:

$$E[N] = 4, \quad \text{Var}(N) = 4.$$

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## Problem 5 – Poisson Approximation to Binomial (Website Clicks)

**Problem.** A website has  $n = 10,000$  visitors/day. Each visitor clicks with probability  $p = 0.0005$ . Let  $X$  be the number of clicks.

### (a) Exact distribution

Number of clicks = number of “successes” in 10,000 independent trials:

$$X \sim \text{Binomial}(n = 10,000, p = 0.0005).$$

### (b) Why Poisson approximation is valid; value of $\lambda$

- $n$  is large (10,000),
- $p$  is small (0.0005),
- $np$  is moderate.

Thus a **Poisson approximation** is appropriate with

$$\lambda = np = 10,000 \cdot 0.0005 = 5.$$

So approximately

$$X \approx \text{Poisson}(\lambda = 5).$$

### (c) Approximate $P(X = 0)$ and $P(X \geq 1)$

Using  $\text{Poisson}(5)$ :

$$P(X = 0) \approx e^{-5} \frac{5^0}{0!} = e^{-5} \approx 0.00674.$$

Then

$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - 0.00674 = 0.99326.$$

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## Problem 6 – Sum of Independent Poisson Variables

**Problem.**  $X_1 \sim \text{Poisson}(2)$  and  $X_2 \sim \text{Poisson}(3)$  are independent. Let  $T = X_1 + X_2$ .

### (a) Distribution of $T$

The sum of independent Poisson random variables is Poisson with parameter equal to the sum of the parameters:

$$T \sim \text{Poisson}(2 + 3) = \text{Poisson}(5).$$

### (b) Compute $P(T = 4)$

Using Poisson(5) pmf:

$$P(T = 4) = e^{-5} \frac{5^4}{4!} = e^{-5} \frac{625}{24}.$$

Numerically:

$$P(T = 4) \approx 0.175.$$

### (c) Mean and variance of $T$

For Poisson(5),

$$E[T] = 5, \quad \text{Var}(T) = 5.$$

Alternatively, using linearity:

$$E[T] = E[X_1] + E[X_2] = 2 + 3 = 5,$$

$$\text{Var}(T) = \text{Var}(X_1) + \text{Var}(X_2) = 2 + 3 = 5,$$

because  $X_1, X_2$  are independent.

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## Problem 7 – Standard Normal $Z \sim N(0, 1)$

Let  $Z$  be standard normal.

### (a) Shape and parameters

- Mean (center): 0
- Variance: 1
- Standard deviation: 1
- Symmetric and bell-shaped about 0.

**(b) Express probabilities using  $\Phi$** 

Let  $\Phi(z) = P(Z \leq z)$  be the standard normal CDF.

$$1. P(Z \leq 1.2) = \Phi(1.2).$$

$$2. P(Z > -0.5) = 1 - P(Z \leq -0.5) = 1 - \Phi(-0.5). \text{ Using symmetry, } \Phi(-0.5) = 1 - \Phi(0.5), \text{ so}$$

$$P(Z > -0.5) = \Phi(0.5).$$

$$3. P(-1 < Z < 2) = P(Z < 2) - P(Z \leq -1) = \Phi(2) - \Phi(-1). \text{ Again using symmetry, } \Phi(-1) = 1 - \Phi(1), \text{ so}$$

$$P(-1 < Z < 2) = \Phi(2) - (1 - \Phi(1)) = \Phi(2) + \Phi(1) - 1.$$

(If numeric values are needed, we can plug in standard normal table values.)

**Problem 8 – Normal with Mean 50 and Variance 9**

Let  $X \sim N(\mu = 50, \sigma^2 = 9)$ . Then  $\sigma = 3$ .

**(a) Standardization**

Define

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 50}{3} \sim N(0, 1).$$

**(b)  $P(X \leq 47)$** 

Convert to  $Z$ :

$$P(X \leq 47) = P\left(Z \leq \frac{47 - 50}{3}\right) = P(Z \leq -1) = \Phi(-1).$$

Using symmetry,  $\Phi(-1) = 1 - \Phi(1)$ , so numerically  $\approx 0.1587$ .

**(c)  $P(46 < X < 55)$** 

Convert both endpoints:

$$P(46 < X < 55) = P\left(\frac{46 - 50}{3} < Z < \frac{55 - 50}{3}\right) = P\left(-\frac{4}{3} < Z < \frac{5}{3}\right).$$

So

$$P(46 < X < 55) = \Phi\left(\frac{5}{3}\right) - \Phi\left(-\frac{4}{3}\right).$$

Using symmetry again,

$$\Phi(-4/3) = 1 - \Phi(4/3).$$

(d)  $P(X > 50)$

Since the distribution is symmetric about 50,

$$P(X > 50) = P(X < 50) = 0.5.$$


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## Problem 9 – Heights $H \sim N(170, 8^2)$

Let  $H$  (height in cm) follow  $N(\mu = 170, \sigma^2 = 64)$ , so  $\sigma = 8$ .

(a) **z-score for 186 cm**

$$z = \frac{186 - 170}{8} = \frac{16}{8} = 2.$$

(b)  $P(H > 186)$

Convert to standard normal:

$$P(H > 186) = P(Z > 2) = 1 - \Phi(2).$$

Numerically this is  $\approx 0.0228$ .

(c) **90th percentile height**

We want  $h$  such that

$$P(H \leq h) = 0.9.$$

Standardize:

$$P\left(Z \leq \frac{h - 170}{8}\right) = 0.9.$$

Let  $z_{0.9}$  be the 0.9-quantile of  $N(0, 1)$ . From tables,  $z_{0.9} \approx 1.28$ . So

$$\frac{h - 170}{8} = z_{0.9} \approx 1.28 \Rightarrow h = 170 + 8(1.28) \approx 170 + 10.24 = 180.24 \text{ cm}.$$


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## Problem 10 – Normal Approximation to Binomial

Let  $X \sim \text{Binomial}(n = 100, p = 0.4)$ .

(a) **Mean and variance**



$$E[X] = np = 100(0.4) = 40.$$

$$\text{Var}(X) = np(1 - p) = 100(0.4)(0.6) = 24.$$

$$\sigma = \sqrt{24} \approx 4.90.$$

**(b) Approximate  $P(X \leq 35)$  with continuity correction**

Use the normal approximation:

$$X \approx N(\mu = 40, \sigma^2 = 24).$$

For  $P(X \leq 35)$ , apply continuity correction:

$$P(X \leq 35) \approx P(Y \leq 35.5)$$

where  $Y \sim N(40, 24)$ .

Standardize:

$$z = \frac{35.5 - 40}{\sqrt{24}} \approx \frac{-4.5}{4.90} \approx -0.92.$$

So

$$P(X \leq 35) \approx \Phi(-0.92).$$

Numerically,  $\Phi(-0.92) \approx 0.18$  (about 18%).

## Problem 11 – Comparing Binomial, Poisson, and Normal

We are given three random variables, each with mean 10: -  $X_1 \sim \text{Binomial}(100, 0.1)$ , -  $X_2 \sim \text{Poisson}(10)$ , -  $X_3 \sim N(10, \sigma^2)$ .

**(a) Compute variances**

1. For  $X_1$ :

$$\text{Var}(X_1) = np(1 - p) = 100(0.1)(0.9) = 9.$$

2. For  $X_2$ :

$$\text{Var}(X_2) = \lambda = 10.$$

**(b) Choose  $\sigma^2$  between these**

We want

$$9 < \sigma^2 < 10.$$

Any choice in this interval works, e.g.

$$\sigma^2 = 9.5.$$

### (c) When is each model appropriate?

- **Binomial** ( $X_1$ ): Use when you have a fixed number  $n$  of independent trials, each with the same success probability  $p$ , and you count the number of successes.
- **Poisson** ( $X_2$ ): Use when events occur randomly and independently over time or space with a constant average rate  $\lambda$ , especially when modeling **rare events** or approximating binomial with large  $n$  and small  $p$ .
- **Normal** ( $X_3$ ): Use for continuous quantities that are sums/averages of many small independent effects (by the central limit theorem), or as an approximation to binomial/other distributions when sample sizes are large and distribution is roughly symmetric.

## Problem 12 – Sample Mean and Central Limit Theorem

Let  $X$  have mean  $\mu$  and variance  $\sigma^2$ . Let  $X_1, X_2, \dots, X_n$  be i.i.d. copies of  $X$ . Define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

**(a) Show**  $E[\bar{X}_n] = \mu$  **and**  $\text{Var}(\bar{X}_n) = \sigma^2/n$

**Expectation:**

$$E[\bar{X}_n] = E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{n\mu}{n} = \mu.$$

**Variance:** (using independence and variance rules)

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

**(b) Why does the variance decrease with  $n$ ?**

Because we are averaging more and more independent observations. The random fluctuations tend to "cancel out," so the spread (variance) of the average becomes smaller. Mathematically, since  $\text{Var}(\bar{X}_n) = \sigma^2/n$ , as  $n$  increases,  $\sigma^2/n$  shrinks toward 0.

### (c) Informal statement of the Central Limit Theorem (CLT)

Informally, the CLT says:

If  $X_1, X_2, \dots, X_n$  are i.i.d. with mean  $\mu$  and variance  $\sigma^2$ , then as  $n$  becomes large, the distribution of the standardized sum

$$> \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} >$$

becomes approximately standard normal  $N(0, 1)$ .

Equivalently, the sample mean  $\bar{X}_n$  is approximately normal with mean  $\mu$  and variance  $\sigma^2/n$  for large  $n$ .

This is why the normal distribution is so widely used for approximating sums and averages, even if the original data are not normally distributed.